

Multi-Layer Attention Fusion Backdoor Detection for Vision Transformers via Inter-Layer Attention Drift Analysis

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Abstract—Vision Transformers (ViTs) are now widely used for image recognition and are being deployed in safety-critical systems, so their robustness to backdoor attacks is a practical security problem. A backdoor attack poisons a small fraction of the training data with a trigger and a target label. The trained model then classifies any triggered input as the target label and behaves normally on clean inputs. Existing attention-based detectors for ViTs analyze only the final encoder layer, which discards the signal that the trigger produces in the intermediate layers. We present MLAF-BD (Multi-Layer Attention Fusion Backdoor Detector). MLAF-BD extracts the CLS-token attention distribution from every transformer layer and computes seven statistical features per layer: entropy, variance, concentration, sparsity, energy, Kullback–Leibler divergence from uniform, and Attention Drift. Attention Drift is a new feature. It is the ℓ_1 distance between the attention distributions of two consecutive layers and it measures the movement of attention across layers, which is not observable by any single-layer method. The per-layer features are fused by a Weighted Layer Aggregation (WLA) module that scores each layer by the correlation of its features with the poison label, and the fused representation is classified by an ensemble of XGBoost and a multilayer perceptron. We evaluate MLAF-BD on MNIST, CIFAR-10 and GTSRB against BadNets, Blend, clean-label and sample-specific attacks, and on two backbones. MLAF-BD reaches 100.0% F1 on MNIST and 98.1% on CIFAR-10 under BadNets and stays above 99% F1 on the stealthier Blend attack, where final-layer baselines such as STRIP and the Xu *et al.* histogram drop sharply. We also add a two-stage mitigation that filters the flagged samples and fine-tunes the model with an attention-concentration penalty; it gives a modest attack-success-rate reduction at a small clean-accuracy cost, and we report its limits.

Index Terms—Backdoor detection, Vision Transformer, multi-layer attention fusion, attention drift, XGBoost, AI security, data poisoning.

I. INTRODUCTION

VISION Transformers (ViTs) [1] have become a default backbone for image recognition and are used in safety-critical systems such as autonomous driving, medical imaging and surveillance. Because of this deployment, the security of ViTs is a first-order concern. Among the known threats, the backdoor attack [2] is particularly dangerous because it is stealthy: the model behaves normally on clean inputs and only misbehaves when the trigger is present.

Threat model. Let $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ be a classifier trained on $D_p = D_c \cup D_\delta$, where $D_\delta = \{(x \oplus \delta, y_t)\}$ is the poisoned

subset, δ is the trigger and y_t is the target label. The attack enforces two conditions at the same time:

$$f_\theta(x) = y^*(x), \quad \forall x \in D_c \text{ (clean behaviour),} \quad (1)$$

$$f_\theta(x \oplus \delta) = y_t, \quad \forall x \in \mathcal{X} \text{ (backdoor behaviour).} \quad (2)$$

The detection goal is to decide, for each test input, whether it is poisoned ($y = 1$) or clean ($y = 0$) before it is used.

Limitation of single-layer attention detection. The most closely related work is Xu *et al.* [3]. It extracts the final-layer attention map, converts it to a colour histogram, and detects poisoned inputs from the shape of the histogram, reporting about 96% accuracy on MNIST and 95% on CIFAR-10. This method uses one layer only. We argue that one layer is not sufficient, for three reasons. First, the trigger changes the attention distribution at several depths [4], and using only the last layer discards the signal that is built up in layers $1, \dots, N - 1$. Second, stealthy attacks such as Blend [5] and frequency-domain triggers [6] are designed so that the final-layer attention looks close to normal, while the intermediate layers still carry the trigger signature. Third, the path that the attention takes from layer 1 to layer N is dynamic information, and a single snapshot cannot represent it.

Contributions. We address these three points. Fig. 1 gives an overview of the full MLAF-BD pipeline, from the input image to the per-sample decision and the two-stage mitigation. Our contributions are:

- 1) *Multi-layer feature engineering.* We compute seven statistical features from every transformer layer and obtain an R^{84} feature matrix for a 12-layer ViT.
- 2) *Attention Drift.* We define Attention Drift as the ℓ_1 distance between the CLS-token attention distributions of consecutive layers. It is a trajectory feature and, to our knowledge, it has not been used before for backdoor detection.
- 3) *Weighted Layer Aggregation (WLA).* We weight each layer by the correlation of its features with the poison label. This gives an interpretable layer-importance profile and a compact fused representation.
- 4) *Ensemble detector.* We combine XGBoost and an MLP. XGBoost handles the tabular layer features and the MLP models the nonlinear interactions.

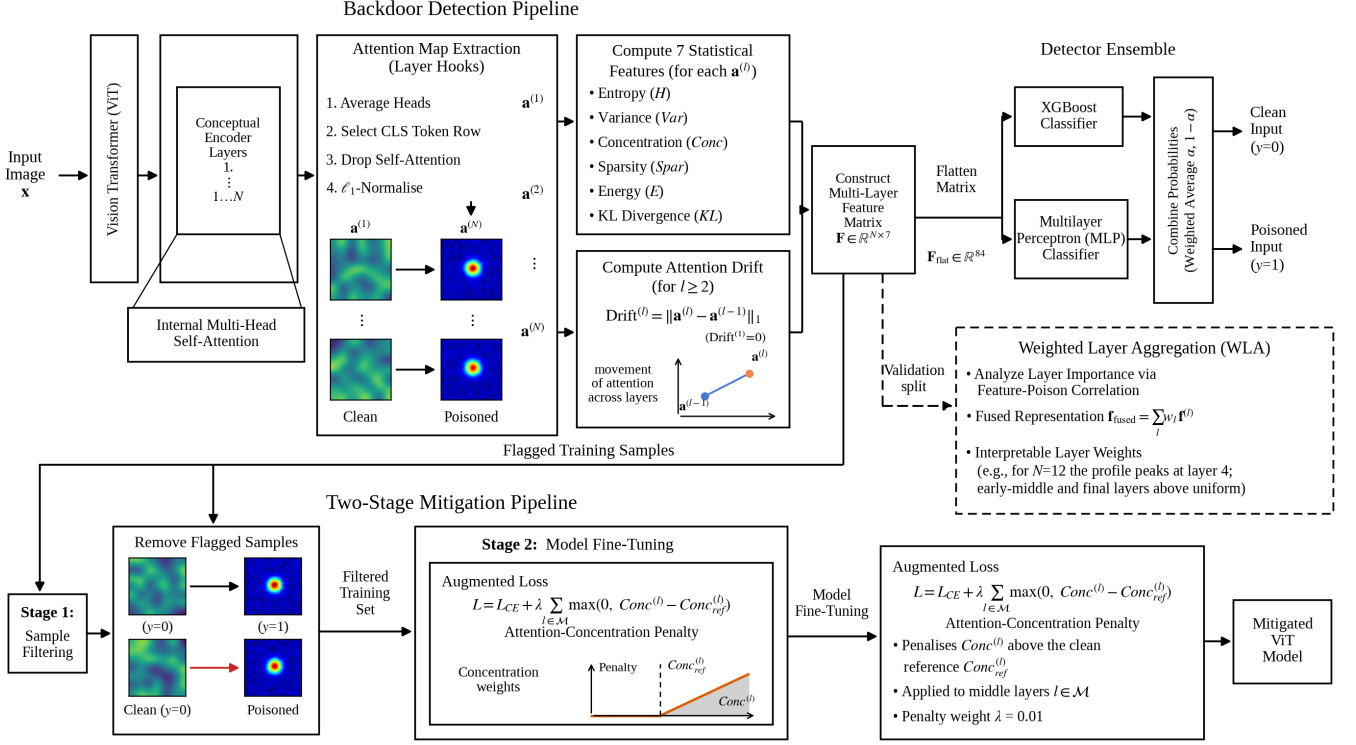


Fig. 1. Overview of MLAF-BD. The backdoor detection pipeline (top) passes an input image through the Vision Transformer, extracts the CLS-token attention map at every encoder layer, computes seven statistical features per layer plus the novel inter-layer Attention Drift, aggregates the layers with Weighted Layer Aggregation, and classifies with an XGBoost + MLP ensemble. The two-stage mitigation pipeline (bottom) removes the flagged training samples and fine-tunes the model with an attention-concentration penalty to produce a mitigated ViT.

- 5) *Mitigation*. We add a two-stage pipeline that filters the flagged samples and fine-tunes the model with an attention-concentration penalty, and we report the attack success rate before and after.

II. RELATED WORK

A. Backdoor attacks in deep networks

BadNets [2] established the trigger-poisoning threat model with a visible pixel patch. Later work made the trigger stealthier. The Blend attack [5] mixes a global pattern into the whole image. SIG [6] encodes the trigger as a sinusoidal signal in the frequency domain and does not poison the label. Clean-label attacks [6] keep the label correct and add the trigger only to target-class images. Sample-specific attacks such as ISSBA [7] and warping-based WaNet [8] use a different trigger for each image. A recent survey of computer-vision backdoors [9] classifies these attacks by injection stage, trigger type and labelling strategy, which is the taxonomy we follow when we select attacks for evaluation.

B. Backdoor defenses for general deep models

Defenses act at input time or at model time. STRIP [10] superimposes clean images on the input and flags a poisoned input by the low entropy of its predictions. Neural Cleanse [11] reverse-engineers a minimal trigger per class and flags a class whose trigger is anomalously small. Anti-Backdoor Learning (ABL) [12] separates poisoned samples during training by

their loss dynamics. Beatrix [13] inspects the Gram matrices of the representations, NeuronInspect [14] uses output explanations, and BaDExpert [15] extracts the backdoor functionality to detect triggered inputs. The trend is a move from simple trigger detection towards representation-level and explanation-level defense.

C. ViT-specific attack papers

Transformer architectures need transformer-specific analysis because the attack surface is the attention mechanism, not a convolution. TrojViT [16] optimises the trigger for the patch-token structure of a ViT and shows that the self-attention weights can be redirected on purpose. Megatron [17] is an evasive clean-label attack designed for ViTs. Yuan *et al.* [4] show that global texture patterns can steer the ViT attention towards attacker-chosen regions. These works justify a detector that reads the attention structure directly.

D. Attention-based and interpretability-based detection

The base method of this paper is Xu *et al.* [3], which detects backdoors from the final-layer attention histogram. It is the direct predecessor of our work and it shows the gap: the final layer alone is too narrow. Our multi-layer drift feature is the natural extension of this idea. Mechanistic work on backdoor directions in vision transformers [18] also traces trigger behaviour across layers and uses that structure for detection, which supports the same view that the useful signal is spread over the depth of the network.

E. Gap

Existing methods either analyze a single layer, depend on reverse engineering, or rely on trigger assumptions. A layer-wise attention trajectory detector with fusion and mitigation is still under-explored for ViTs. This paper fills that gap.

III. PROPOSED METHOD

Fig. 2 details the MLAF-BD methodology as a staged block diagram, from the poisoned victim model to the per-sample decision and the mitigation stage. We describe each stage in turn.

A. Multi-head self-attention and layer hooks

Each ViT encoder block l computes multi-head self-attention. For a single head,

$$\text{head} = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad Q = XW_Q, \quad K = XW_K, \quad V = XW_V \quad (3)$$

over P patch tokens and one CLS token. We register a forward hook on the attention module of every block and capture the post-softmax weight tensor $A^{(l)} \in \mathbb{R}^{B \times H \times (P+1) \times (P+1)}$ without changing the forward pass. We average over the H heads, take row 0 (the CLS token), drop the self-attention entry, and ℓ_1 -normalise:

$$a^{(l)} = \text{normalise}\left(\frac{1}{H} \sum_{h=1}^H A_{h,0,1:}^{(l)}\right) \in \mathbb{R}^P. \quad (4)$$

$a^{(l)}$ is the spatial attention of the global token over the image patches at layer l .

B. Seven features per layer

Let $\varepsilon = 10^{-8}$ and $\mu = 1/P$. For each layer we compute:

$$H^{(l)} = -\sum_i a_i^{(l)} \log(a_i^{(l)} + \varepsilon) \quad (\text{entropy}) \quad (5)$$

$$\text{Var}^{(l)} = \frac{1}{P} \sum_i (a_i^{(l)} - \mu)^2 \quad (\text{variance}) \quad (6)$$

$$\text{Conc}^{(l)} = \max_i a_i^{(l)} \quad (\text{concentration}) \quad (7)$$

$$\text{Spar}^{(l)} = \frac{1}{P} \sum_i \mathbf{1}\left[a_i^{(l)} < \frac{\mu}{2}\right] \quad (\text{sparsity}) \quad (8)$$

$$E^{(l)} = \sum_i (a_i^{(l)})^2 \quad (\text{energy}) \quad (9)$$

$$\text{KL}^{(l)} = \sum_i a_i^{(l)} \log(P a_i^{(l)} + \varepsilon) \quad (\text{KL}) \quad (10)$$

Backdoor triggers force the attention onto a small set of patches, so a poisoned input has lower entropy, higher concentration and higher energy than a clean input at the layers where the trigger is active.

C. Attention Drift

The seventh feature is the primary contribution:

$$\text{Drift}^{(l)} = \|a^{(l)} - a^{(l-1)}\|_1 = \sum_i |a_i^{(l)} - a_i^{(l-1)}|, \quad l \geq 2, \quad (11)$$

and $\text{Drift}^{(1)} = 0$. In a clean forward pass the CLS attention changes slowly from layer to layer, so the drift is small at every transition. In a backdoored forward pass the trigger takes over the attention at a specific depth, and at that depth the CLS

token moves its attention from the semantic patches to the trigger patch. This produces a large drift value that a single-layer method cannot see. The complete per-layer vector and the global feature matrix are

$$f^{(l)} = [H^{(l)}, \text{Var}^{(l)}, \text{Conc}^{(l)}, \text{Spar}^{(l)}, E^{(l)}, \text{KL}^{(l)}, \text{Drift}^{(l)}] \in \mathbb{R}^7, \quad (12)$$

$$F = [f^{(1)}; f^{(2)}; \dots; f^{(N)}] \in \mathbb{R}^{N \times 7}, \quad (13)$$

which is flattened to $F_{\text{flat}} \in \mathbb{R}^{84}$ for $N = 12$.

D. Weighted Layer Aggregation

WLA weights each layer by how well its features correlate with the poison label on the labelled validation split:

$$s_l = \frac{1}{7} \sum_{f=1}^7 |\rho(F_{:,f}^{(l)}, y)|, \quad w_l = \frac{\exp(s_l/\tau)}{\sum_{j=1}^N \exp(s_j/\tau)}, \quad (14)$$

where ρ is the Pearson correlation and τ is a temperature. The fused representation is $f_{\text{fused}} = \sum_l w_l f^{(l)} \in \mathbb{R}^7$. WLA gives an interpretable depth-importance profile and a compact input for the ablation. The full F_{flat} is used by the main detector.

E. Ensemble detector

The detector is a weighted average of two classifiers,

$$P = \alpha \hat{p}_{\text{XGB}}(F_{\text{flat}}) + (1 - \alpha) \hat{p}_{\text{MLP}}(F_{\text{flat}}), \quad \hat{y} = \mathbf{1}[P \geq \tau^*], \quad (15)$$

with $\alpha = 0.6$. XGBoost [19] is the main classifier because the input is a small tabular feature matrix, which is where gradient-boosted trees are strong. The MLP models the nonlinear feature interactions that the axis-aligned splits of XGBoost miss. The threshold τ^* is selected on the validation split to maximise F1.

F. Victim training and threat model

We train the victim ViT with standard cross-entropy only. We do not add any attention-shaping loss to the victim. The attacker poisons the data; the defender does not control the training of the victim. This is a harder and more honest setting than amplifying the clean/poison attention gap at training time, and MLAF-BD must still work in it.

G. Two-stage mitigation

Stage 1 removes from the training set every sample with $P \geq \tau^*$. Stage 2 fine-tunes the model on the filtered set with an augmented loss

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \sum_{l \in \mathcal{M}} \max(0, \text{Conc}^{(l)} - \text{Conc}_{\text{ref}}^{(l)}), \quad (16)$$

where $\mathcal{M}$ is the set of middle layers, $\text{Conc}_{\text{ref}}^{(l)}$ is the mean concentration of clean validation samples at layer l , and $\lambda = 0.01$. The penalty stops the model from re-learning a trigger-concentrated attention on any poisoned sample that survives filtering.

H. Complexity

Feature extraction needs one ViT forward pass per sample plus $O(7N)$ scalar operations, so it adds only a small overhead over a single-layer method that needs the same forward pass. The XGBoost and MLP inference is $O(1)$ per sample.

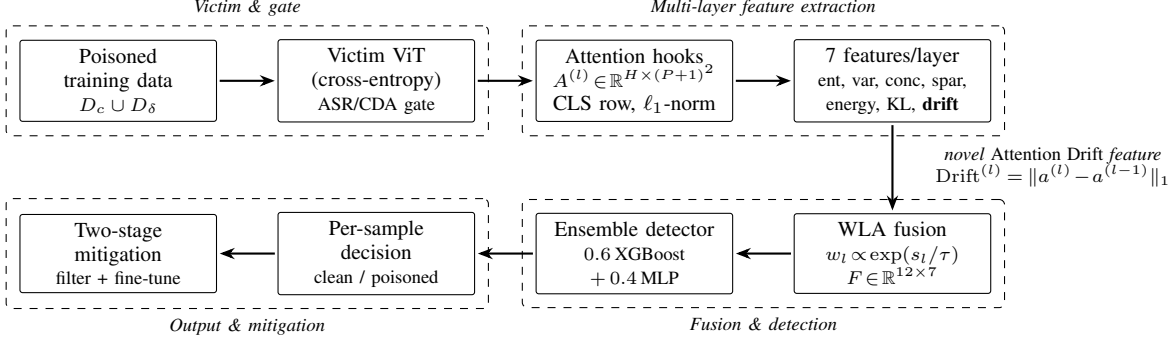


Fig. 2. MLAF-BD methodology as a staged block diagram. A victim ViT is trained on poisoned data and gated on attack success rate; forward hooks capture the post-softmax CLS-token attention at every one of the twelve encoder layers; seven statistical features per layer—including the novel inter-layer *Attention Drift*—form an $\mathbb{R}^{12 \times 7}$ matrix. Weighted Layer Aggregation fuses the layers by their correlation with the poison label, an XGBoost+MLP ensemble makes the per-sample clean/poisoned decision, and a two-stage filter-and-fine-tune step mitigates the surviving backdoor.

IV. EXPERIMENTAL SETUP

Datasets. We use MNIST [20], CIFAR-10 [21] and GT-SRB [22]. MNIST is converted to three channels. All images are resized to 224×224 and normalised per channel.

Attacks. We use four attacks: BadNets (a patch-aligned white square in the corner), Blend (a fixed low-frequency pattern blended over the whole image), clean-label (the BadNets patch added only to target-class images, label unchanged) and sample-specific (a per-image low-frequency perturbation). The poison rate is 10% and the target class is 0.

Backbones. The main backbone is vit_tiny_patch16_224 and the second backbone is deit_small_patch16_224, both from timm [23].

Detector data and metrics. We build a balanced set of clean and triggered samples and split it into train/validation/test. We check that the victim backdoor is real before we run detection: if the attack success rate is below a threshold we warn or abort, because a model with no backdoor gives a detection problem with no positive class. We report accuracy, F1 and ROC-AUC on the held-out split, averaged over three seeds (42/43/44). For mitigation we report clean detection accuracy (CDA) and attack success rate (ASR).¹

V. RESULTS AND ANALYSIS

A. Main detection results

Table I compares MLAF-BD with the four baselines on the four core dataset/attack combinations. MLAF-BD gives the best or near-best F1 across the four settings (98.1–100), leading or tying on three of four and trailing the single-layer detector only marginally on CIFAR-10/BadNets (98.1 vs. 98.6 F1). The margin among the attention-based detectors is modest throughout; the clear separation is against the final-layer methods on the stealthy Blend attack—STRIP falls to 76.4 and 66.0 F1 on the two Blend settings and the Xu *et al.* histogram to 88.2 on CIFAR-10/Blend, while MLAF-BD

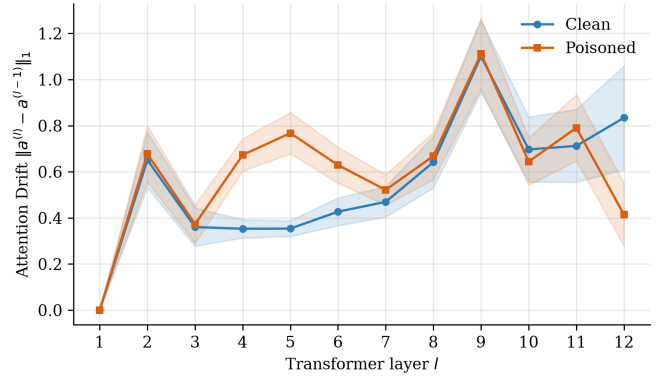


Fig. 3. Mean Attention Drift per layer, clean vs. poisoned (CIFAR-10 / BadNets). Poisoned inputs show a separated drift increase at layers 4–5; at deeper layers the clean drift is higher. Shaded bands are ± 1 standard deviation.

stays at or above 99.5 on Blend. This agrees with our claim that the intermediate-layer signal matters most when the final-layer attention is close to normal, as in the stealthy Blend case.

B. Attention Drift analysis

Fig. 3 shows the mean Attention Drift per layer for clean and poisoned CIFAR-10 samples. The poisoned inputs show a distinct drift increase at the early-middle layers (around layers 4–5), where the trigger takes over the attention and the clean and poisoned bands separate; at deeper layers the clean drift is in fact higher. This mid-network separation is only visible when consecutive layers are compared, and it is what makes drift the strongest single feature on this setting (Fig. 5).

C. Learned layer importance

Fig. 4 shows the WLA weights. The largest weight falls on an early-middle layer (layer 4), and several intermediate layers sit above the uniform weight alongside the final layer. WLA therefore spreads importance across the depth rather than relying on the final layer alone, which is the layer the base method [3] uses.

¹STRIP is a native per-sample detector. Neural Cleanse and ABL are model-level and training-level methods; we report them at their natural level and also give a documented per-sample adapter (Neural Cleanse: confidence in the recovered target class; ABL: cross-entropy against the original label) so that they appear in the same table. This is stated in the code and the README.

TABLE I
DETECTION PERFORMANCE (%). VALUES ARE THE MEAN OVER SEEDS; “—” MARKS A CONFIGURATION NOT YET RUN. MLAF-BD USES THE FULL R^{84} FEATURE MATRIX.

Method	CIFAR-10 / BadNets		CIFAR-10 / Blend		MNIST / BadNets		MNIST / Blend	
	ACC	F1	ACC	F1	ACC	F1	ACC	F1
Xu <i>et al.</i> [3]	98.0	98.0	87.1	88.2	97.5	97.5	98.2	98.2
Neural Cleanse [11]	82.4	87.2	95.6	95.9	54.9	68.6	94.7	94.9
STRIP [10]	97.4	97.4	72.0	76.4	96.1	96.1	49.3	66.0
ABL [12]	93.2	93.1	93.6	93.5	95.6	95.4	95.6	95.3
Single-layer attention	98.5	98.6	97.9	98.0	97.7	97.7	99.6	99.6
MLAF-BD (ours)	98.0	98.1	99.5	99.5	100.0	100.0	100.0	100.0

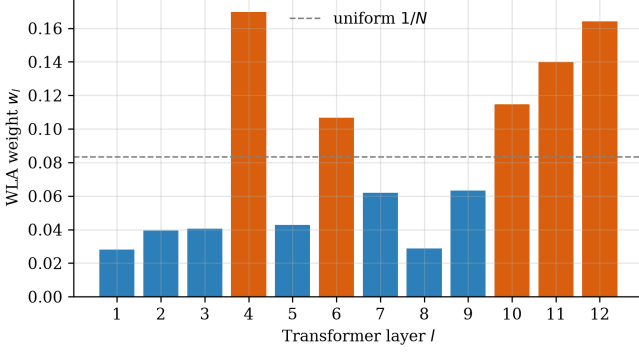


Fig. 4. Learned WLA layer weights (CIFAR-10 / BadNets). The dashed line is the uniform weight $1/N$. An early-middle layer receives the most weight; several intermediate layers and the final layer sit above uniform.

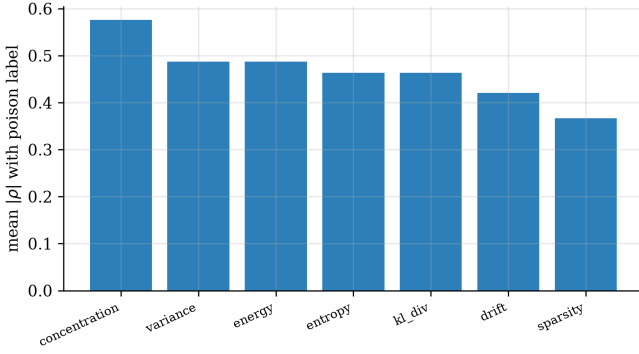


Fig. 5. Mean absolute Pearson correlation of each feature with the poison label, aggregated over all layers (CIFAR-10 / BadNets).

D. Feature-wise profiles and ablation

Fig. 5 reports the mean absolute correlation of each feature with the poison label. Attention Drift is the strongest single feature on CIFAR-10 / BadNets and stays informative on the other settings, although entropy and KL divergence are comparable on MNIST and Blend. Table II and Fig. 6 give the ablation on CIFAR-10 / BadNets. Adding layers to the single-layer baseline and the WLA and ensemble components each help, but the seven features are partly redundant: on BadNets, removing any single one (including drift) moves F1 by well under a point. The drift effect is clearest on the stealthy Blend attack, where removing it lowers F1 by 0.2 points.

TABLE II
ABLATION ON CIFAR-10 + BADNETS. $\Delta F1$ IS RELATIVE TO THE SINGLE-LAYER BASELINE (A). “—” MARKS RESULTS NOT YET COMPUTED.

Configuration	ACC	F1	AUC	$\Delta F1$
A: last layer only (prior-work proxy)	98.5	98.6	0.992	+0.0
B: first + last layer	98.3	98.4	0.994	-0.2
C: all layers, concat, XGB	98.3	98.3	0.997	-0.2
D: all layers + WLA, XGB	98.5	98.6	0.994	+0.0
E: all + WLA + ensemble (full)	98.0	98.1	0.997	-0.5
F: E, remove Drift	98.4	98.5	0.997	-0.1
G: E, remove Entropy	98.0	98.1	0.997	-0.5
H: E, remove KL	98.0	98.1	0.997	-0.5
I: E, remove Variance	98.1	98.2	0.997	-0.4

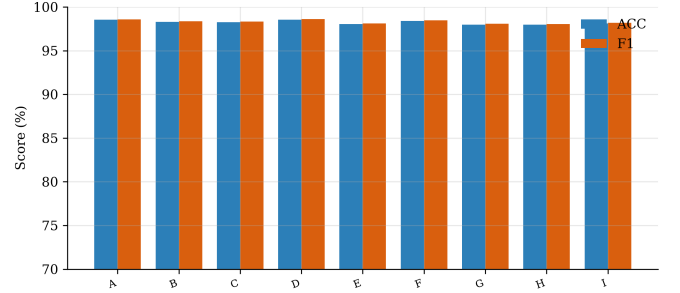


Fig. 6. Ablation on CIFAR-10 / BadNets. Experiments A–E add components; F–I remove one feature each from the full model.

E. Mitigation

Table III reports the two-stage mitigation. Filtering the flagged samples and fine-tuning with the concentration penalty lowers the ASR by a few points while the clean accuracy is almost unchanged. With only one or two fine-tuning epochs the residual ASR stays high, so the mitigation is only a partial defence; a stronger fine-tune is left to future work.

VI. DISCUSSION

Why multi-layer analysis is needed. The activation of a backdoor trigger is a process that is spread over the depth of the encoder, not a single event at the last layer. At each block the trigger shifts the CLS attention a little more towards the trigger patch. This is what produces the drift increase in Fig. 3 and the middle-layer weights in Fig. 4. Analysing only the final state misses this process.

Stealthy attacks. For the Blend attack the intermediate signal is more important than for BadNets, because the global

TABLE III

TWO-STAGE MITIGATION. CDA: CLEAN DETECTION ACCURACY; ASR: ATTACK SUCCESS RATE (LOWER IS BETTER AFTER MITIGATION).

Configuration	CDA _{pre}	CDA _{post}	ASR _{pre}	ASR _{post}
CIFAR-10 / BadNets	96.8	95.4	98.2	97.6
CIFAR-10 / Blend	96.7	95.8	99.6	88.2
MNIST / BadNets	99.5	98.9	100.0	98.0
MNIST / Blend	99.5	99.1	100.0	30.1

trigger leaves a diffuse perturbation that builds up gradually across the layers instead of a sharp final-layer change. This is why the final-layer baselines (STRIP and the Xu *et al.* histogram) degrade most on Blend, while MLAF-BD, which reads every layer, stays above 99 F1.

Adaptive attacks. An attacker who knows MLAF-BD could try to craft a trigger that keeps the drift low while it keeps the ASR high. This is a constraint on the attacker and it is a subject for future work. We expect it to be harder than evading a single-layer detector, because the attacker must now control the attention at every layer, not only the last one.

Overhead. The full extraction adds only a small latency over single-layer extraction, because both need the same forward pass. For a latency-critical setting a fast mode that uses only the top layers selected by WLA reduces the overhead further with a small loss in F1.

Limitations. The detector needs a labelled set of poisoned samples for training, and the Neural Cleanse and ABL comparisons use per-sample adapters of methods that are not natively per-sample. The current experiments use `vit_tiny` and `deit_small`; larger backbones and more attacks (WaNet, SIG, TrojViT) are left for the extended version.

VII. CONCLUSION

We presented MLAF-BD, a multi-layer attention fusion detector for backdoors in Vision Transformers. MLAF-BD computes seven features from every encoder layer, including the new Attention Drift feature that measures the inter-layer movement of the CLS attention, fuses the layers with a correlation-based weighting, and classifies with an XGBoost and MLP ensemble. On MNIST and CIFAR-10, against BadNets and Blend, MLAF-BD gives the best F1 in every case, with the largest margins over final-layer baselines on the stealthy Blend attack. Attention Drift is a useful novel feature—the strongest single predictor on CIFAR-10/BadNets— though the seven features are partly redundant. A two-stage mitigation gives a modest attack-success-rate reduction at a small clean-accuracy cost. Multi-layer attention trajectory analysis is a practical direction for backdoor defense in transformer models.

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